

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the Interstate	)	Docket No. RM26-4-000
Transmission System	)	

COMMENTS OF FIRSTENERGY SERVICE COMPANY

Pursuant to the Federal Energy Regulatory Commission’s (“Commission”) October 27, 2025, Notice Inviting Comments on the Secretary of Energy’s proposed advance notice of proposed rulemaking (“ANOPR”) requested under section 403 of the Department of Energy Reorganization Act¹ on October 23, 2025, and the Commission’s Notice Granting Extension of Time on November 7, 2025, FirstEnergy Service Company (“FirstEnergy”)² respectfully submits the following comments.³

I. BACKGROUND

FirstEnergy’s electric distribution and transmission companies form one of the nation’s largest investor-owned electric systems, serving more than six million customers in Ohio, Pennsylvania, New Jersey, West Virginia, Maryland and New York. FirstEnergy’s transmission owning subsidiaries operate more than 24,000 miles of transmission lines that connect the Midwest and Mid-Atlantic regions.

¹ 42 U.S.C. § 7173.

² FirstEnergy also submits these comments on behalf of its transmission-owner affiliates American Transmission Systems, Incorporated, Jersey Central Power & Light Company, Mid-Atlantic Interstate Transmission LLC, The Potomac Edison Company, Monongahela Power Company, Keystone Appalachian Transmission Company, and Trans-Allegheny Interstate Line Company.

³ FirstEnergy is also joining comments submitted jointly with certain other PJM transmission owners. FirstEnergy encourages the Commission to read these comments in conjunction with those.

I. EXECUTIVE SUMMARY

On September 18, 2025, the Department of Energy’s (the “Department”) Grid Deployment Office issued its Speed to Power Initiative to accelerate the development of transmission and generation in the United States to bring the wires and power to market that are necessary to winning the global artificial intelligence race.⁴ FirstEnergy shares the goals of the Administration and is devoting its capital and resources to ensuring large load data center customers are “able to connect to the transmission system in a timely, orderly, and non-discriminatory manner.”⁵

If the Department and Commission desire large load data center interconnections to proceed at a quicker pace and in a more orderly manner, different processes are generally required for large loads than the interconnection procedures used by electric generation. Unlike generation facilities, large load data center customers plan to consume power rather than to generate and to dispatch power. Accordingly, data centers do not require open access to the interstate transmission system to export power. Rather, they not only require reliable electric delivery and supply service at the point of interconnection like other load customers, but given their size and operational characteristics they require more consideration when interconnecting to the system.

If the Commission elects to promulgate a rule for the interconnection of large load data center interconnections, it should do so in a targeted manner.

First, to the extent the Commission elects to work toward a final rule, it should define a “large load customer” to mean a data center customer whose facility is at least 200 MW. Doing so will prevent the final rule from frustrating the goals of the ANOPR by adversely affecting

⁴ Department of Energy, *Energy Department Launches Speed to Power Initiative, Accelerating Large-Scale Grid Infrastructure Projects* (Sept. 18, 2025), <https://www.energy.gov/articles/energy-department-launches-speed-power-initiative-accelerating-large-scale-grid>.

⁵ Letter from Secretary Chris Wright to Commissioners of FERC, dated Oct. 23, 2025 (“Secretary’s Letter”), at 1.

industrial, manufacturing and other large load customers who are not the intended targets of the rule. An overly broad definition would undermine community growth, hamper efforts to reshore manufacturing, and stall—if not reverse—job growth, all of which are Administration priorities.

Second, the Commission should consider adopting interconnection study and interconnection queue constructs that allow qualified customers to move quickly through the queue ahead of non-viable projects.

Third, the Commission should continue to rely on its long-standing policy of rolling the costs of integrated network upgrades into network transmission rates.

Fourth, the Commission should abandon any effort to provide load customers an option to build as the proposal raises reliability concerns and would undercut rather than promote efficiency.

Finally, regardless of how it proceeds on individual aspects of the proposed ANOPR, the Commission should continue to keep reliability front of mind in this rulemaking process. Each of these measures will help achieve the Department’s speed to power goals. Limiting the scope of any final rule will help mitigate unintended consequences that may result from a broader, more complicated rule that actually slows the pace of large load interconnections and has adverse consequences for non-data center load customers.

II. TARGETED GUIDANCE ON LARGE LOAD INTERCONNECTION WILL HELP ACCELERATE THE DEPARTMENT’S SPEED TO POWER GOALS

A. The Class of Customers Affected by the Proposed ANOPR Should be Limited to Data Center Customers with Loads of no Less Than 200 MW

The ANOPR proposes to regulate the interconnections of large loads that require electric power capacities of 20 MW and higher.⁶ However, it is primarily aimed at data center load, which typically is far larger than 20 MW. To ensure that any rule adopted by the Commission is

⁶ ANOPR at 1, 10-11.

appropriately targeted, and does not frustrate the goals of the ANOPR, the Commission should limit any rule to (1) data centers that (2) have loads of at least 200 MW.

Sweeping in all large loads of 20 MW and above risks clogging interconnection queues, creating long interconnection wait times, and threatening the expedient interconnection of new data center load. Studying different types of load customers that have different operational attributes in a single queue will create unnecessary complexity and reduce opportunities for streamlining and other efficiencies. A one-size-fits-all approach also risks undermining the collaborative processes by which FirstEnergy has been able to work with customers of varying sizes and configurations in order to identify the best location for interconnecting new facilities. Imposing a 20 MW large load interconnection threshold also may result in prohibitory costs and onerous regulatory hurdles for other types of load customers, harming economic development that relies on the efficient and cost-effective interconnection of manufacturers and other traditional large load customers. Such an outcome would work contrary to the goal of revitalizing the United States' manufacturing base and the associated beneficial job creation and positive community impact. For these reasons, the goals of the ANOPR, would be best met by limiting any Commission-mandated interconnection procedures to data center loads of 200 MW or greater in size.⁷

B. The Commission Should Take Steps to Facilitate the Speed of any Large Load Interconnection Queue

In the Secretary of Energy's letter accompanying the ANOPR, the Secretary recognized that timely and efficient interconnection of large customer loads is critical to maintaining system

⁷ As discussed below, a threshold that is too low, coupled with allowing an option to build, will result in transmission owners, such as FirstEnergy, being required to expend significant additional resources overseeing customer-built facilities, detracting from important grid planning and management activities, to the detriment of customers.

reliability.⁸ For this objective to be realized, however, any interconnection queue and study processes adopted for large data center customers must be structured to prioritize projects that are ready to proceed.

FirstEnergy supports an interconnection framework that promotes the efficient, timely and reliable interconnection of large loads. Consistent with the Department’s objectives in the ANOPR and the Commission’s mandate under the Federal Power Act (“FPA”), the interconnection framework adopted in this proceeding should incorporate processes and lessons learned from the generator interconnection process that facilitate the timely processing of large interconnection requests through the queue.

The Commission should consider reforms in this proceeding that allow interconnection-ready projects to timely advance through the queue, similar to PJM Interconnection, L.L.C.’s (“PJM”) consideration of rules for an expedited interconnection track for the interconnection of data centers and other large loads.⁹ By allowing projects that meet defined technical, financial, and permitting milestones to proceed ahead of other less-developed requests, transmission providers can reduce unnecessary delays and lower the queue backlog. This type of reform aligns with the Commission’s broader policy goals articulated in Order No. 2023, which emphasizes clustering and milestone-based prioritization to facilitate timely interconnection.¹⁰

⁸ Secretary’s Letter at 1.

⁹ See PJM, *Large Load Additions PJM Conceptual Proposal and Request for Member Feedback* (Aug. 18, 2025), <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20250818/20250818-item-03---pjm-conceptual-proposal-and-request-for-member-feedback---presentation.pdf>; see also PJM, *Large Load Additions CIFP Update* (Oct. 1, 2025), <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20251001/20251001-item-04---cifp---lla-updates---pjm-presentation.pdf>.

¹⁰ *Improvements to Generator Interconnection Procs. and Agreements*, Order No. 2023, 184 FERC ¶ 61,054, *order on reh’g*, 185 FERC ¶ 61,063 (2023), *order on reh’g*, Order No. 2023-A, 186 FERC ¶ 61,199 (2024), *pet. for review pending sub nom. Advanced Energy United v. FERC*, Nos. 23-1282 *et al.* (D.C. Cir.).

Experience over the past two decades has demonstrated that congested interconnection queues are among the greatest barriers to achieving fast and efficient interconnections. One of the primary reasons that interconnection queues have stagnated is the prevalence of speculative projects and the relative ease, even with increased financial commitments, with which developers can remain in the queue for long periods even after a project's viability is reduced. Provisions that allow more viable "shovel ready" projects to proceed, while ensuring that access to the queue remains open and non-discriminatory, are key to ensuring the success of any set of large load data center interconnection procedures that the Commission adopts in this proceeding.

III. THE DEPARTMENT'S PROPOSED "DIRECT ASSIGNMENT" RATES TREATMENT MAY CAUSE SUBSTANTIAL DELAY OF THE PROCESSING OF LARGE LOAD DATA CENTER CUSTOMER INTERCONNECTION REQUESTS, THEREBY FRUSTRATING THE GOAL OF SPEED TO POWER

FirstEnergy recognizes the ANOPR's goal to achieve procedures that facilitate the efficient, timely, and reliable interconnection of large load customers. But the ANOPR's suggestion that "load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies"¹¹ threatens to inhibit this goal. The proposed direct assignment of network upgrade costs to an end-use customer raises thorny jurisdictional issues that will result in years of delay, thereby frustrating efficient and timely data center interconnections. This is because, by definition, almost all data center customers are retail customers, and it is the state regulator and not FERC that establishes retail rates. Any attempt to go beyond FERC's established jurisdiction over transmission rates¹² could result in years of delay

¹¹ ANOPR at 10.

¹² See *New York v. FERC*, 535 U.S. 1, 17, 28 (2002) (quoting 16 U.S.C. § 824(b)).¹³ See *Otter Tail Power Co.*, Opinion No. 93, 12 FERC ¶ 61,169, at 61,420 (1980); see also *Idaho Power Co.*, Opinion No. 13, 3 FERC ¶ 61,108 (1978); *Utah Power & Light Co.*, Opinion No. 220, 27 FERC ¶ 61,258, *reh'g denied*, 28 FERC ¶ 61,088 (1984), *aff'd sub nom. Sierra Pac. Power Co. v. FERC*, 793 F.2d 1086 (9th Cir. 1986); *Niagara Mohawk Power Corp.*, Opinion No. 296, 42 FERC ¶ 61,143 (1988).

as jurisdictional challenges are sorted out in the appellate courts. Rather than risk extended delay due to jurisdictional challenges and other cost allocation battles, the Commission should employ its long-standing rolled-in rate approach for network upgrades in conjunction with facilitating customer protections to ensure that other retail load customers are not adversely affected by large load interconnections.

There is no need to risk such delay. The Commission's well-established rolled-in rate treatment, coupled with customer protections like those already being implemented by FirstEnergy and other transmission owners and their load-serving entity affiliates, provide an effective avenue to address large load interconnections. The Commission has long maintained a policy in favor of socializing transmission costs rather than directly assigning those costs to individual customers, especially when the network upgrades benefit the broader grid or regional reliability.¹³ Over the years, the Commission has consistently found that transmission upgrades integrated with the larger network benefit the entire system. The Commission has found that costs applicable to integrated transmission facilities that are necessary for the connection of a load customer should not be directly assigned.¹⁴

FirstEnergy recognizes the importance of protecting existing load customers from any adverse impacts of data center interconnections. However, there is no need to deviate from longstanding Commission policy and treat the allocation of network upgrade costs driven by large load data center customers differently to do so. FirstEnergy affiliates that are wholesale

¹³ See *Otter Tail Power Co.*, Opinion No. 93, 12 FERC ¶ 61,169, at 61,420 (1980); see also *Idaho Power Co.*, Opinion No. 13, 3 FERC ¶ 61,108 (1978); *Utah Power & Light Co.*, Opinion No. 220, 27 FERC ¶ 61,258, *reh'g denied*, 28 FERC ¶ 61,088 (1984), *aff'd sub nom. Sierra Pac. Power Co. v. FERC*, 793 F.2d 1086 (9th Cir. 1986); *Niagara Mohawk Power Corp.*, Opinion No. 296, 42 FERC ¶ 61,143 (1988).

¹⁴ See *Duke Energy Carolinas, LLC*, 168 FERC ¶ 61,190, at P 18 (2019) (citing *Ne. Tex. Elec. Coop.*, 108 FERC ¶ 61,084 (2004); *Am. Elec. Power Serv. Corp.*, 44 FERC ¶ 61,206 (1988), *reh'g denied*, 45 FERC ¶ 61,408 (1988), *reh'g denied*, 46 FERC ¶ 61,382 (1992)).

transmission customers/load serving entities are already implementing mechanisms to uphold cost-causation principles and to protect customers from the impacts of large load interconnections. These mechanisms are built on collective experience and collaboration with regulators across jurisdictions and demonstrate that large load interconnections can be accomplished quickly and efficiently while protecting other retail customers from bearing unwarranted costs.

Such protections can be incorporated into agreements between utilities and large load customers or retail tariffs to ensure that costs are appropriately allocated and that large load customers actually pay the costs they are allocated based on the amount of load that they claim they will bring to the system. In short, rather than wading into complex jurisdictional issues and other cost allocation disputes likely involved in a direct assignment approach, the Commission should use its traditional rolled-in rate approach to ensure that wholesale transmission customers pay their fair share of network upgrade costs that benefit all customers, and the Commission should allow state regulators and load-serving entities to use tariff and/or agreement provisions at the retail level to ensure that large load customers pay their fair share of retail-level costs.

IV. IMPLEMENTATION AND ADMINISTRATION OF AN OPTION TO BUILD WILL DELAY THE INTERCONNECTION OF LARGE LOAD DATA CENTER CUSTOMERS

The ninth principle of the ANOPR proposes an option to build for large load customers that would ostensibly be the same as that afforded to generator interconnection customers pursuant to Order No. 845.¹⁵ Following the promulgation of the regulations resulting from Order No. 845,

¹⁵ ANOPR at P 26; *see also Reform of Generator Interconnection Procs. & Agreements*, Order No. 845, 163 FERC ¶ 61,043 (2018), *order on reh'g*, Order No. 845-A, 166 FERC ¶ 61,137, *order on reh'g*, Order No. 845-B, 168 FERC ¶ 61,092 (2019); *Standardization of Small Generator Interconnection Agreements & Procs.*, Order No. 2006, 111 FERC ¶ 61,220, at PP 546-547, *order on reh'g*, Order No. 2006-A, 113 FERC ¶ 61,195 (2005), *order granting clarification*, Order No. 2006-B, 116 FERC ¶ 61,046 (2006).

transmission providers implemented an option to build for generator interconnection customers¹⁶ that permits such customers to design, procure, and construct Transmission Provider's Interconnection Facilities¹⁷ and Stand Alone Network Upgrades.¹⁸

FirstEnergy views the option to build for generator interconnection customers as a failed tariff mechanism that has not resulted in meaningful value or savings that were expected by independent power producers. It is a challenging tool for customers to implement, and in most cases, it slows down the engineering design of interconnection facilities because several sets of engineers from various entities have to review design plans and determine whether the facilities are appropriately constructed.

However, regardless of whether the Commission believes that an option to build is still a valuable tool for generator interconnection customers, it is not appropriate to extend it to large load data center customers. Implementation and administration of an option to build for large load data center customers will add significant delay and risk to their proposed interconnections. Although they are sophisticated business entities, large load customers are not jurisdictional public utilities and do not have the necessary experience to safely and cost-efficiently design and

¹⁶ *Pro forma* LGIA, Article 5.1.3 (Option to Build).

¹⁷ Transmission Provider's Interconnection Facilities are defined as all facilities and equipment owned, controlled, or operated by Transmission Provider from the Point of Change of Ownership to the Point of Interconnection as identified in Appendix A to the Standard Large Generator Interconnection Agreement, including any modifications, additions or upgrades to such facilities and equipment. Transmission Provider's Interconnection Facilities are sole use facilities and shall not include Distribution Upgrades, Stand Alone Network Upgrades or Network Upgrades. *Pro forma* LGIA, (Definitions).

¹⁸ Stand Alone Network Upgrades are defined as Network Upgrades that are not part of an Affected System that Interconnection Customer may construct without affecting day-to-day operations of the Transmission System during their construction and the following conditions are met: (1) a Substation Network Upgrade must only be required for a single Interconnection Customer in the Cluster and no other Interconnection Customer in that Cluster is required to interconnect to the same Substation Network Upgrades, and (2) a System Network Upgrade must only be required for a single Interconnection Customer in the Cluster, as indicated under Transmission Provider's Proportional Impact Method. *Pro forma* LGIA, (Definitions).

construct facilities that meet reliability standards. Allowing an option to build, particularly if the Commission opts for an overly expansive definition of large load customers, will result in transmission owners such as FirstEnergy expending significant additional labor and equipment resources to properly oversee customer-built projects to ensure that they are built to utility standards and subject to utility commissioning requirements. This will detract from other priorities relating to planning and operating the grid, to the detriment of other customers.

A. Transmission Providers are Better Equipped to Obtain State Siting Authority to Construct Interconnection Facilities and Network Upgrades

Giving large load customers the option to build is more likely to delay large load customer interconnections than to accelerate them. Large load customers would need to acquire adequate expertise before even starting the construction process which includes skills applicable to designing, planning, securing land acquisitions, and permitting and siting, and some of these functions may still need to be managed by the transmission-owning utility. For instance, permitting and siting are often lengthy, complicated processes that require strong regulatory expertise in a number of areas, as well as experience managing community outreach and other stakeholder interactions.

Thus, it is likely that the transmission utility will need to continue to manage state siting and permitting applications and resulting proceedings. In order to do so in the context of a large load customer option to build would require developing a complex web of interactions between the large load customer, the utility and regulators that will likely result in significant delays in siting and permitting proceedings. Providing large load customers an option to build will therefore frustrate the ANOPR's goal of accelerating the interconnection of large loads. The most efficient and cost-effective approach is to have experienced transmission-owning utilities continue to build the transmission facilities associated with large load customers.

B. There Are Other Serious Impediments to an Option to Build Such That the ANOPR’s Goals of Efficiency and Speed To Power Will be Best Realized by Having Transmission-Owning Utilities Construct Upgrades

Adopting the option to build for large load customers is apt to create problems rather than solutions. If option to build is invoked, the development and construction of large load interconnection facilities risk falling behind schedule due to concerns with the customer’s ability to comply with North American Electric Reliability Corporation (“NERC”) standards. Transmission owner oversight of customers that elect the option to build may lead to disputes over whether the customer is in compliance with NERC Reliability Standards and the transmission owner’s design standards.

As end-use customers, large load customers have no obligations to protect or maintain the reliability of the transmission system, including compliance with applicable NERC standards.¹⁹ Instead, it is the transmission-owning public utilities that will continue to have the obligation to ensure compliance with NERC standards relating to such things as facility ratings, protection systems, and clearances. Providing load customers an option to build will put the transmission owner in an untenable position whereby it has compliance responsibilities for facilities that it did not engineer or build. This is not just and reasonable; transmission owners should not be held responsible for the reliability performance of network upgrades designed and built by others.

¹⁹ Load, as defined by NERC, is not itself an applicable entity. Load is defined by NERC as “[a]n end-use device or customer that receives power from the electric system.” See NERC, *Glossary of Terms Used in NERC Reliability Standards* (Oct. 1, 2025), https://www.nerc.com/pa/Stand/Glossary%20of%20Terms/Glossary_of_Terms.pdf, at 24. A load is only subject to mandatory NERC reliability standards if it meets the criteria to register as a user, owner, or operator of the Bulk Electric System. Loads that do not qualify for registration therefore have no direct compliance obligations under NERC standards. Moreover, NERC currently does not have a finalized definition for “large load” that would trigger compliance with its reliability standards. Although the ANOPR states that NERC should review its reliability standards to determine if categories or modified standards are necessary, this directive provides no assurances that large load customers will be subject to any relevant NERC obligations in the future, nor if so, when. ANOPR at P 31.

Similarly, large load customers are not familiar with Regional Transmission Organization and transmission owner planning processes, which have to be navigated thoroughly to develop interconnection facilities and network upgrades. Transmission owners must comply with outage coordination requirements and other stakeholder process requirements before constructing and electrifying facilities. These are not processes that large load customers have experience with, and it makes little sense to burden customers with those requirements.

Electric utilities, using existing expertise, tools and processes, are best positioned to efficiently and cost-effectively construct necessary network transmission upgrades driven by large load interconnections. Transmission-owning public utilities, which are legally obligated to ensure system safety and reliability, have the appropriate incentive to build and to maintain network upgrades properly. Constructing transmission to accommodate load is the very essence of a transmission-owning utility's business, and it makes little sense that customers in an entirely different line of business would be better equipped, or have the appropriate incentives, to construct transmission network upgrades in a more efficient or cost-effective manner.

Even if the Commission believes that large load customers should have some option to build, any such option would need to be subject to a number of conditions in order to ensure that the applicable transmission upgrades are constructed safely and reliably. First, any option to build should be limited solely to transmission interconnection facilities and stand-alone network upgrades, such as the customer side of the substation where service drops from the transmission grid to customer-specific facilities. Under no circumstances should an option to build extend to energized facilities, or facilities beyond the point of interconnection, such as work on integrated, networked transmission lines.

Importantly, large load customers should be required to strictly adhere to the design and materials/equipment standards of the interconnecting transmission owner. Failure to enforce consistent design standards would inject additional uncertainties into the planning process, complicating the operations, communications, and systems required for reliable operations and potentially degrading the reliability of the system. A large load customer should also be required to utilize contractors approved by the transmission utility.

This leads full circle back to the fact that the best way to achieve the ANOPR's goal of timely large load interconnections is to have experienced transmission-owning utilities design and build the transmission facilities needed to interconnect large load customers. FirstEnergy and other transmission-owning utilities have the requisite knowledge and experience to develop and to construct large load interconnection facilities and network upgrades quickly and reliably. It is the fabric of the transmission business to (1) serve customers and (2) invest in the growth and reliability of the system to benefit society and shareholders. An option to build, which in theory is to benefit users of the system that possess no operations experience, is in practice disruptive to those core public utility functions and should be discarded as a concept in this rulemaking discussion.

V. CONCLUSION

FirstEnergy appreciates this opportunity to comment on the proposed ANOPR. It is further encouraged to assist the Department and the Commission in their efforts to ensuring large load data center customers have the opportunity to connect to the grid quickly and economically. It is important that large load data center customers pay their share of system upgrades while other load customers are allocated a portion of those costs in a manner that is at least roughly commensurate with the benefits they receive from such upgrades. FirstEnergy looks forward to collaborating with the Commission, other transmission owners, states, consumer advocates, and customers to help achieve the Department's vision of establishing a just and reasonable construct that effectuates speed to power for large load data center customers.

Respectfully submitted,

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